

## Point, Line and Edge Detection

Point, Line, and Edge Detection are important techniques in **Digital Image Processing** used to identify significant changes in image intensity.

They help in:

- object detection
  - boundary identification
  - image segmentation
  - feature extraction
- 

### 1. Point Detection

#### Definition

Point detection identifies isolated pixels whose intensity values are significantly different from neighboring pixels.

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#### Example

A bright pixel in a dark area:

```
0 0 0
0 9 0
0 0 0
```

Here, the center pixel is very different.

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#### Method

Usually detected using a mask/filter:

```
-1 -1 -1
-1 8 -1
-1 -1 -1
```

If output exceeds threshold → point detected.

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#### Applications

- Detecting stars in satellite images

- Noise spot detection
  - Defect detection
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## 2. Line Detection

### Definition

Line detection identifies horizontal, vertical, or diagonal lines in an image.

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### Types of Lines

- Horizontal
  - Vertical
  - $+45^\circ$  diagonal
  - $-45^\circ$  diagonal
- 

### Example Masks

#### Horizontal Line Mask

```
-1 -1 -1  
2 2 2  
-1 -1 -1
```

#### Vertical Line Mask

```
-1 2 -1  
-1 2 -1  
-1 2 -1
```

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### Applications

- Road detection
  - Fingerprint analysis
  - Text line detection
- 

## 3. Edge Detection

## Definition

Edge detection identifies boundaries where image intensity changes sharply.

Edges represent object boundaries.

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## Main Idea

If neighboring pixels have large intensity differences, an edge exists.

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## Types of Edges

- Horizontal edge
  - Vertical edge
  - Diagonal edge
- 

## Common Edge Detection Operators

### 1. Sobel Operator

Uses gradient masks.

#### Horizontal Mask

```
-1 -2 -1
0 0 0
1 2 1
```

#### Vertical Mask

```
-1 0 1
-2 0 2
-1 0 1
```

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### 2. Prewitt Operator

Similar to Sobel but simpler.

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### 3. Roberts Operator

Uses 2×2 masks.

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#### 4. Canny Edge Detector

Most advanced and accurate method.

Steps:

1. Noise reduction
2. Gradient calculation
3. Non-maximum suppression
4. Thresholding

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#### Difference Between Point, Line and Edge Detection

| Technique       | Detects           |
|-----------------|-------------------|
| Point Detection | Isolated pixels   |
| Line Detection  | Straight lines    |
| Edge Detection  | Object boundaries |

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#### Applications

Used in:

- Medical imaging
- Face detection
- Object recognition
- Autonomous vehicles
- Computer vision

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#### One-Line Interview Answer

Point detection identifies isolated pixels, line detection identifies straight lines, and edge detection identifies sharp intensity changes representing object boundaries in an image.

## Thresholding

Thresholding is an image segmentation technique used to separate objects from the background by converting a grayscale image into a binary image.

It is one of the simplest and most widely used segmentation methods in Digital Image Processing.

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### Simple Definition

Thresholding divides image pixels into:

- foreground (object)
- background

based on a threshold intensity value.

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### Main Idea

If pixel intensity is:

- greater than threshold → object
  - smaller than threshold → background
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### Mathematical Representation

If threshold value =  $T$

$$g(x, y) = \begin{cases} 1, & f(x, y) \geq T \\ 0, & f(x, y) < T \end{cases}$$

Where:

- $f(x, y)$  = original image
  - $g(x, y)$  = thresholded image
  - 1 = white/object
  - 0 = black/background
- 

### Example

Suppose threshold:

**T = 100**

**Pixel values:**

**50 120 180 90**

**After thresholding:**

**0 1 1 0**

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## **Types of Thresholding**

### **1. Global Thresholding**

**Uses one fixed threshold for the entire image.**

**Example:**

**T = 128**

**Simple and fast.**

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### **2. Local (Adaptive) Thresholding**

**Different threshold values are used for different image regions.**

**Useful when lighting is uneven.**

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### **3. Otsu's Thresholding**

**Automatically calculates the optimal threshold value by minimizing intra-class variance.**

**Very popular automatic method.**

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### **4. Binary Thresholding**

**Produces only:**

- **black**
- **white**

**image.**

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## **Advantages**

- **Simple implementation**
  - **Fast processing**
  - **Effective for high-contrast images**
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## **Disadvantages**

- **Sensitive to noise**
  - **Poor performance with uneven illumination**
  - **Difficult when object and background intensities overlap**
- 

## **Applications**

### **Used in:**

- **OCR (text recognition)**
- **Medical imaging**
- **Object detection**
- **Industrial inspection**
- **Face recognition preprocessing**

## Hough Transform

Hough Transform is a feature extraction technique used to detect geometric shapes such as:

- straight lines
- circles
- curves

in an image.

It is especially useful when shapes are:

- broken
- noisy
- partially hidden

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### Simple Definition

Hough Transform detects shapes by converting image points into a parameter space and finding patterns there.

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### Main Idea

Instead of detecting lines directly in the image space, Hough Transform maps image points into another space called the Hough parameter space.

If many points belong to the same line, their curves intersect at a common point in parameter space.

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### Line Detection Using Hough Transform

A line can be represented as:

$$y = mx + c$$

$m$

$b$

-10-8-6-4-2246810-10-5510y-interceptx-intercept

But this form has problems for vertical lines.



So Hough Transform uses the polar form:

$$\rho = x \cos \theta + y \sin \theta$$

Where:

- $\rho$  = perpendicular distance from origin
  - $\theta$  = angle of the line
- 

## Working Process

### Step 1: Detect Edge Points

Usually using:

- Canny edge detector
  - Sobel operator
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### Step 2: Convert to Parameter Space

Each edge point votes for possible:

- $\rho$
- $\theta$

values.

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### Step 3: Voting Process

An accumulator matrix stores votes.

If many points vote for the same parameters:

- a line exists.
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### Step 4: Detect Peaks

High vote counts indicate detected shapes.

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## Circle Detection

For circles:

$$(x - a)^2 + (y - b)^2 = r^2$$

$h$

$k$

$r$

$$(x)^2 + (y)^2 = 3.0^2$$

-10-8-6-4-2246810-6-4-2246

Where:

- (a,b) = center
- r = radius

Hough Transform can also detect circles using this equation.

#### Advantages

- Detects incomplete lines/shapes
- Robust against noise
- Works for multiple shapes

#### Disadvantages

- Computationally expensive
- Requires large memory
- Slower for complex images

#### Applications

Used in:

- Lane detection in self-driving cars
- Medical imaging
- Shape recognition
- Industrial inspection
- Computer vision

## Region-Based Segmentation

Region-Based Segmentation is an image segmentation technique that divides an image into regions based on similarity in:

- intensity
- color
- texture
- other properties

Pixels with similar characteristics are grouped together to form meaningful regions.

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### Simple Definition

Region-based segmentation groups neighboring pixels that have similar properties into one region.

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### Main Idea

Instead of detecting boundaries directly, this method focuses on finding connected areas with similar features.

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### Characteristics of a Region

A valid region should:

- have similar pixel properties
  - be connected
  - differ from neighboring regions
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## Types of Region-Based Segmentation

### 1. Region Growing

Starts from selected seed points and adds neighboring pixels that satisfy similarity conditions.

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### Steps

1. Select seed pixel
  2. Check neighboring pixels
  3. Add similar pixels to region
  4. Repeat until no more pixels satisfy condition
- 

### **Example**

If neighboring pixels have similar intensity:

50 51 49 52

they are grouped into one region.

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### **2. Region Splitting**

The image is divided into smaller regions until each region becomes homogeneous.

Usually uses quadtree decomposition.

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### **3. Region Merging**

Adjacent similar regions are merged together.

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### **4. Split-and-Merge Method**

Combination of:

- region splitting
- region merging

Improves segmentation accuracy.

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### **Similarity Criteria**

Regions are formed based on:

- gray level similarity
- texture similarity
- color similarity

- statistical properties
- 

### Advantages

- Produces connected regions
  - Effective for noisy images
  - Better object representation
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### Disadvantages

- Sensitive to seed selection
  - Computationally expensive
  - May cause over-segmentation
- 

### Applications

Used in:

- Medical image analysis
  - Satellite image processing
  - Face recognition
  - Object tracking
  - Industrial inspection
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### Region-Based vs Edge-Based Segmentation

| Feature           | Region-Based | Edge-Based       |
|-------------------|--------------|------------------|
| Method            | Similarity   | Intensity change |
| Output            | Regions      | Boundaries       |
| Noise Sensitivity | Lower        | Higher           |

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## Edge and Boundary Linking

Edge and Boundary Linking is a process in Digital Image Processing used to connect detected edge pixels into meaningful boundaries or object shapes.

After edge detection, edges may appear broken or disconnected.

Boundary linking helps form continuous object boundaries.

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### Simple Definition

Boundary linking connects nearby edge pixels that belong to the same object.

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### Main Idea

Detected edges often contain:

- gaps
- noise
- discontinuities

Linking methods combine related edge pixels based on:

- similarity
  - distance
  - direction
- 

### Difference Between Edge and Boundary

| Term | Meaning |
|------|---------|
|------|---------|

|      |                        |
|------|------------------------|
| Edge | Sharp intensity change |
|------|------------------------|

|          |                                |
|----------|--------------------------------|
| Boundary | Complete outer shape of object |
|----------|--------------------------------|

An edge is a local feature, while a boundary forms a complete object outline.

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### Methods of Edge and Boundary Linking

#### 1. Local Processing Method

Edges are linked using local properties like:

- gradient magnitude
- gradient direction

Nearby pixels with similar properties are connected.

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### Example

If neighboring pixels have:

- similar intensity change
- same edge direction

they are linked together.

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## 2. Hough Transform

Used to detect and connect:

- straight lines
- curves
- circles

It converts image points into parameter space.

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### Line Equation

$$\rho = x \cos \theta + y \sin \theta$$

Where:

- $\rho$  = distance from origin
  - $\theta$  = angle
- 

## 3. Graph-Based Linking

Edge pixels are treated as graph nodes and connected based on similarity.

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## 4. Contour Following

**Starts from one edge pixel and follows connected neighboring pixels to create a complete boundary.**

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### **Steps in Boundary Linking**

- 1. Detect edges**
  - 2. Remove noise**
  - 3. Analyze neighboring edge pixels**
  - 4. Connect related edges**
  - 5. Form complete boundaries**
- 

### **Applications**

**Used in:**

- Object recognition**
  - Medical imaging**
  - Face detection**
  - Shape analysis**
  - Autonomous vehicles**
- 

### **Advantages**

- Creates meaningful object shapes**
  - Improves segmentation accuracy**
  - Helps in object detection**
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### **Disadvantages**

- Sensitive to noise**
  - Broken edges may cause incorrect linking**
  - Complex for irregular shapes**
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